

Track Generation Report

=====

Generated: 2026-05-30 12:47:59

Gate file: ref_gates_gates.csv

Gate count: 6

Optimization parameters

method: L-BFGS-B

maxiter: 10

ftol: 1e-07

Point counts

objective function points: 30

path generation points: 30

Results

total optimization time: 1.151 s

total path length: 195.9472 m

Per-step time estimate (one measured objective evaluation)

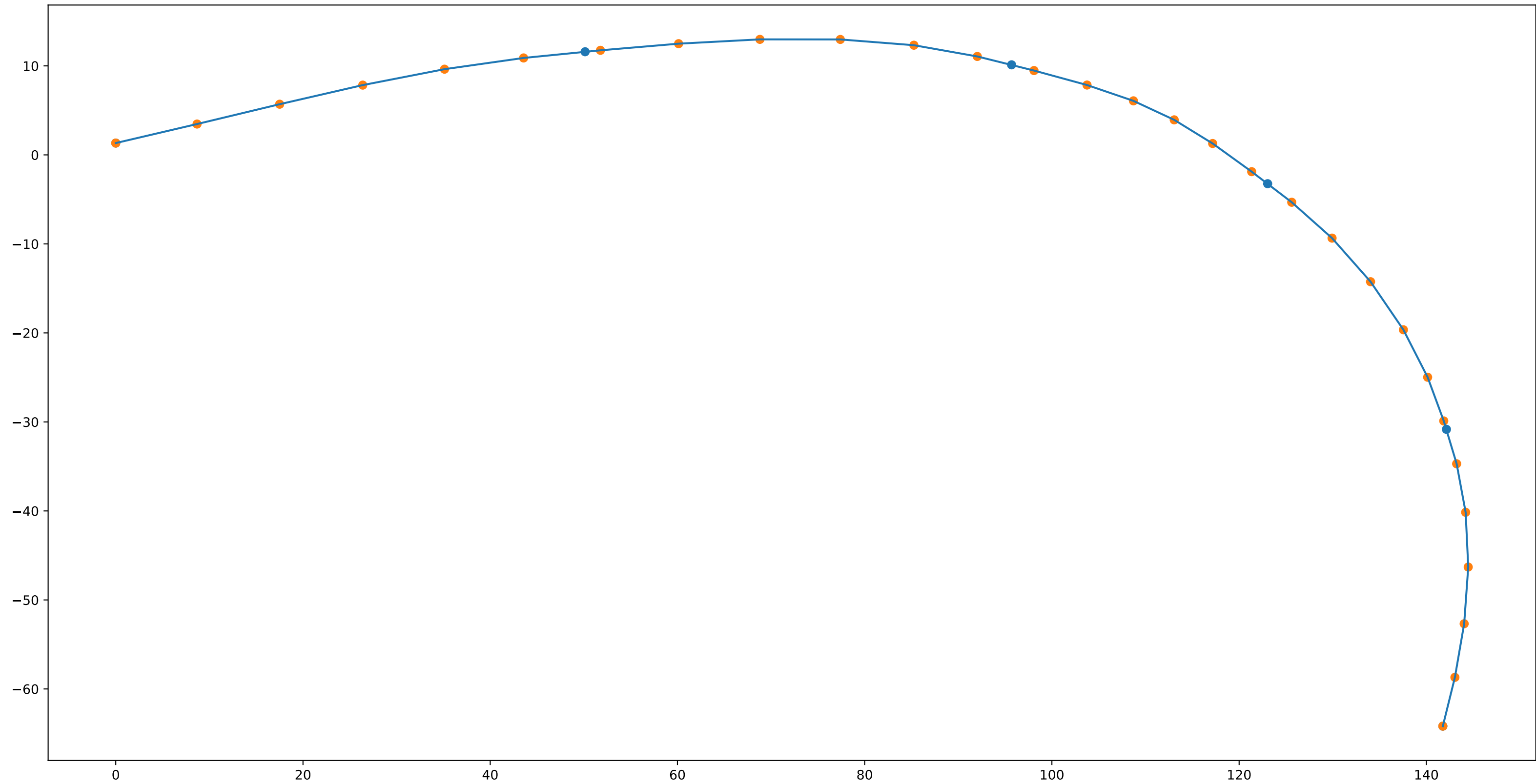
make spline from dofs: 0.79 ms

path length function: 0.58 ms

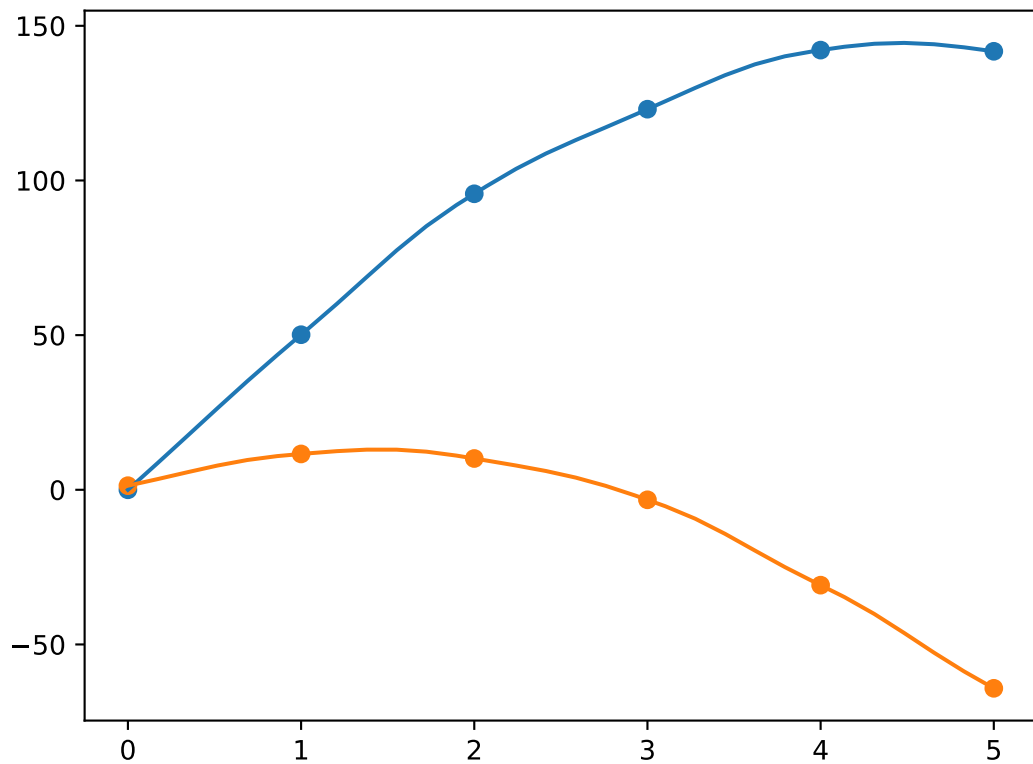
equidistant times generated: 0.53 ms

curvature objective function: 2.70 ms

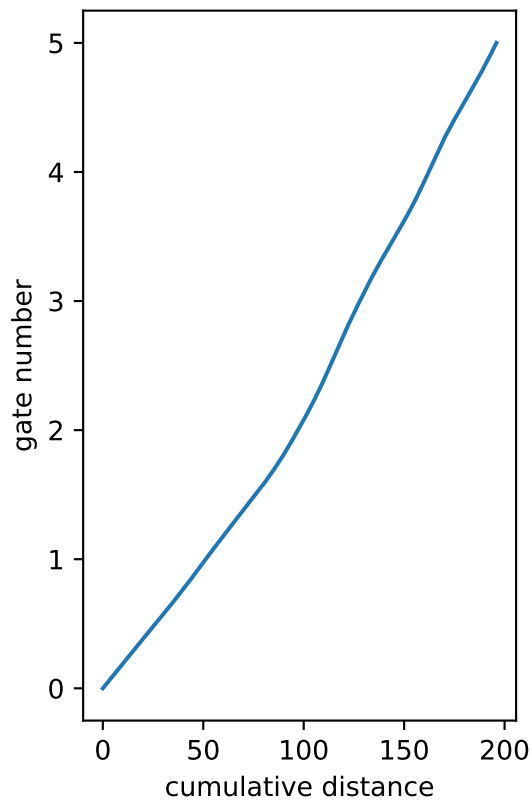
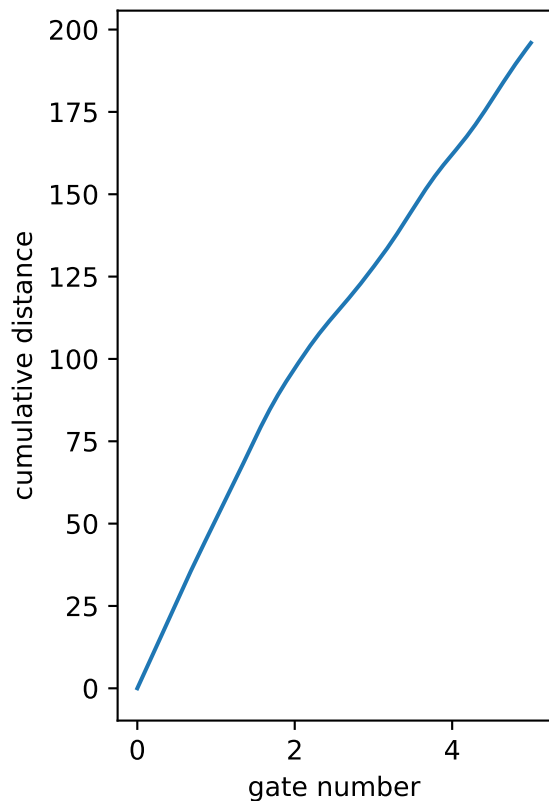
Spline debug: path and gate cross points



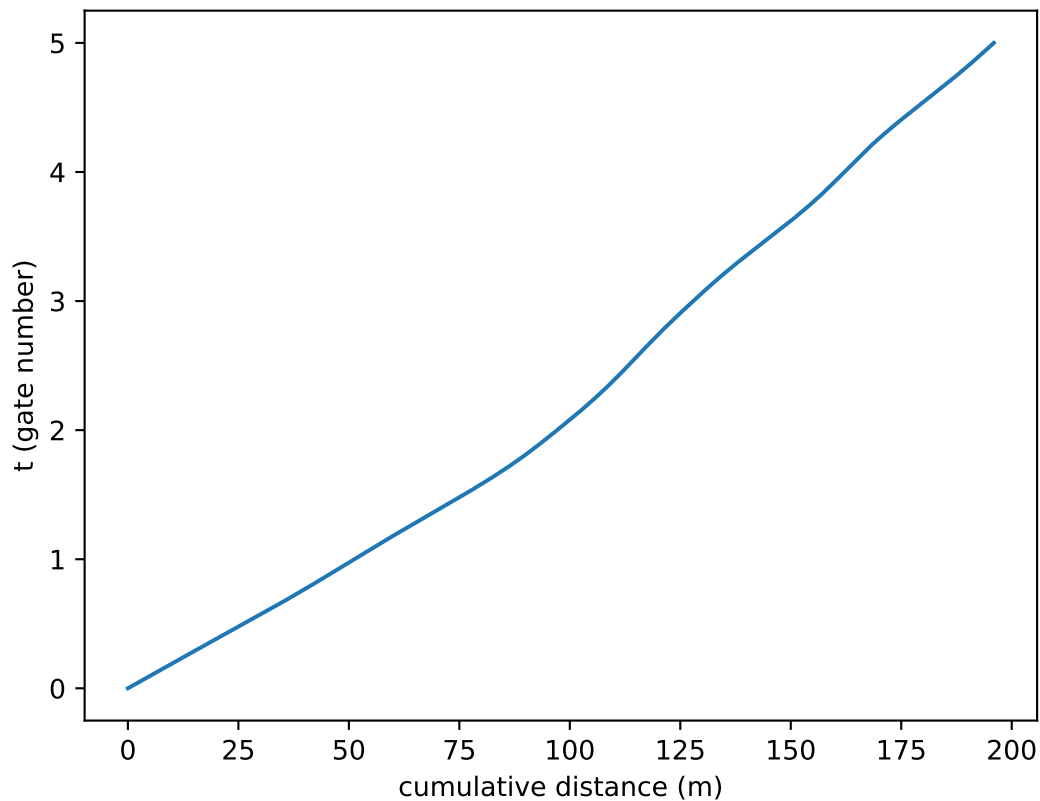
Spline debug: $x(t)$ and $y(t)$



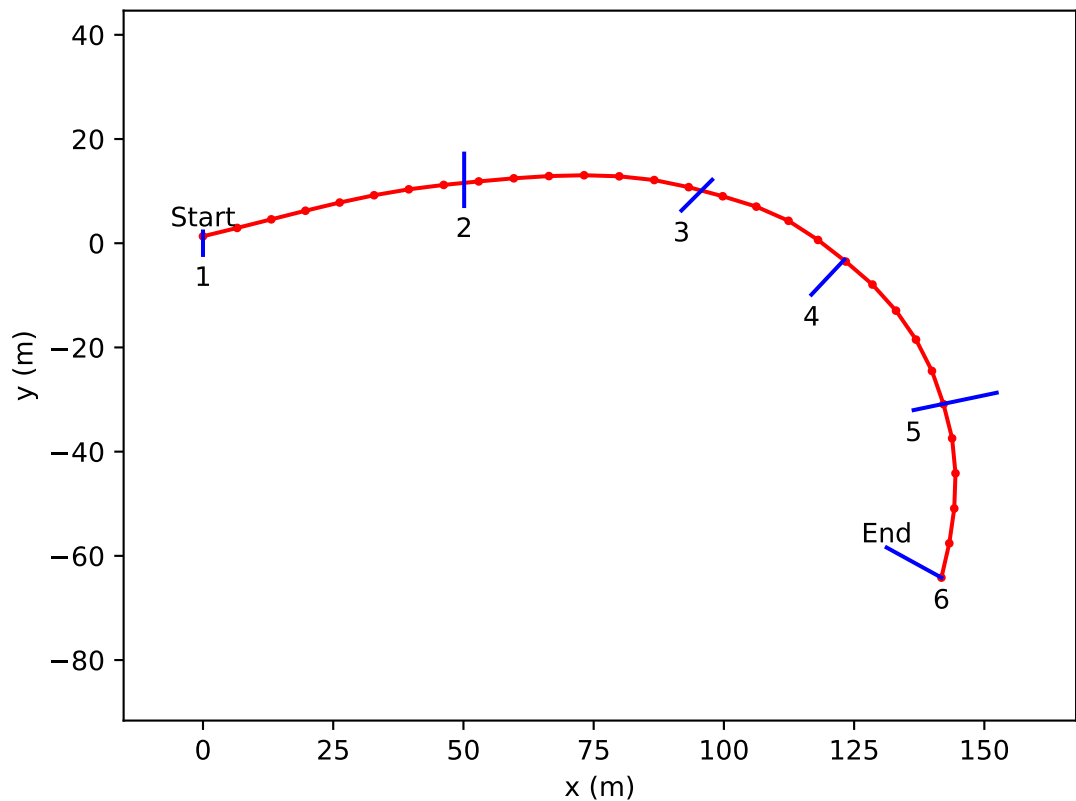
Path debug: cumulative distance vs gate number



Equidistant fit: distance to t mapping



Map plot: optimized path through gates



Optimization iteration overlay

Optimization progression (11 iterations) 5 equally spaced iterations

